Unit 3 – Solid Rocket Engine

Explanatory notes + solved numericals

21ASC305T - Rocket Propulsion
Slides ke basis par simple Hinglish me samjhaya hua notes

Unit 3 - Solid Rocket Engine

Is unit me solid rocket motor ki poori kahani hai: propellant kaisa hota hai, kaise jalta hai, kaise ignite hota hai, burn rate kis cheez se badalta hai, aur last me numericals. Har topic pehle simple language me samjhaya hai, phir exam ke liye main points alag se diye hain.

Is PDF me kya kya hai

- Solid propellant ke 4 types (double-base, composite, CMDB, nitramine)
- Internal ballistics, burning rate vs regression rate
- Burn rate badhane ke tareeke (composite ke liye 5 points)
- Grain design ki 5 requirements
- Ignition process: 3 phases + 10 ignitability factors
- Igniter types: pyrotechnic vs pyrogen, safe and arm device
- Mass flow relation, $p_1 = K \rho_b r C^*$, r = a p^n
- Temperature sensitivity (σ_p, π_K), erosive burning
- Last me 4 numericals poore solve kiye hue

1. Solid propellants - basics aur types

Solid rocket motor me fuel aur oxidizer dono ek hi solid block (grain) me pehle se mile hote hain. Is block ko jalate hain to hot gas banti hai aur nozzle se nikal kar thrust deti hai. Slides ke hisaab se solid propellants 4 category me bante hain:

(a) Double-base ya homogeneous propellant

- Isme mainly 2 energetic cheezein hoti hain: **Nitrocellulose (NC)** aur **Nitroglycerin (NG)**.
- NC fuel-rich component hai, aur NG ek energetic plasticizer hai (matlab NG grain ko soft/processable bhi banata hai aur energy bhi deta hai).
- Isme alag se solid oxidizer ke particles nahi hote jaise composite me hote hain. Poora material ek hi jaisa (homogeneous) hota hai.
- Example: purane gun aur rocket propellants zyadatar double-base concept par bane hain.

(b) Composite ya heterogeneous propellant

- Isme alag-alag physical ingredients hote hain jo alag phases me rehte hain, mix hokar ek nahi ho jaate.
- **Ammonium perchlorate (AP)** = oxidizer, solid particles ki form me binder me bikhra hota hai.
- **Polymer binder** = fuel bhi aur binder bhi (ingredients ko chipka ke rakhta hai).
- **Aluminium** = extra fuel (kuch formulations me), aur thode additives.

(c) Composite Modified Double-Base (CMDB)

- Double-base aur composite dono ka mix. Matlab double-base matrix ke andar solid oxidizer ke particles daal dete hain.

(d) Nitramine propellant

- High-energy propellant jisme energetic material ka bada hissa nitramine compound hota hai (group: N-NO2).
- Common nitramines: **RDX** (Research Department Explosive), **HMX** (High Melting Explosive), **CL-20** (aur zyada energetic, special formulations me).
- Inme nitrogen aur oxygen-rich groups dono hote hain, isliye ye jaldi decompose hote hain aur bahut saari hot gas nikalti hai.

Type	Main ingredients	Ek line me pehchaan
Double-base	NC + NG	Homogeneous, alag oxidizer particle nahi

Type	Main ingredients	Ek line me pehchaan
Composite	AP + polymer binder + Al	Oxidizer particles binder me bikhre hue
CMDB	Double-base + oxidizer particles	Dono ka mix
Nitramine	RDX / HMX / CL-20	N-NO2 group, high energy

2. Internal ballistics - burning rate aur burning surface

Rocket motor ka design aur uska kaam teen cheezon par depend karta hai: **burning rate**, **burning surface** aur **grain geometry**. In teeno ko jo branch study karti hai use **internal ballistics** kehte hain.

Propellant ki burning surface hamesha apne perpendicular direction me andar ki taraf peeche hoti jaati hai. Ye peeche hone ki speed cm/s, mm/s ya in./s me naapte hain, aur ise hi **burning rate r** kehte hain.

Term	Matlab
Burning rate	Solid propellant ki chemical reaction kitni speed se material ko consume kar rahi hai (mm/s ya in./s).
Regression rate	Burning surface kitni speed se grain ke andar perpendicular peeche hoti ja rahi hai jaise grain gas me badal raha hai.

Burning surface contours (slide wala diagram)

Slide me ek 2D grain dikhaya hai jisme central cylindrical cavity me 5 slots hain. Burning surface ki alag-alag position ki lines (contours) draw ki hain, har line ke beech fixed chhota time interval hai. Is diagram se ye pata chalta hai:

- Agar in contour lines ki lengths lagbhag same hain (±15% ke andar), to burning area **constant** maana jaata hai.
- End ke paas burning surface area Ab bahut kam ho jaata hai. Isse thrust aur chamber pressure gir jaate hain.
- Agar pressure ek minimum value ke neeche aa jaye to combustion band ho jaata hai aur kuch propellant bina jale reh jaata hai. Ise **slivers** kehte hain. (Slide me ise 'conflagration value' likha hai, books me ise deflagration limit bolte hain.)
- Rocket motor design ke liye zaroori hai ki propellant ka burn rate saari operating conditions me pata ho.

3. Burning rate kaise badhta ya badalta hai

Kisi bhi propellant ke liye burn rate in cheezon se badhta hai:

- **Start se pehle propellant ka temperature zyada ho** - garam grain jaldi jalta hai.
- **Chamber pressure zyada ho.**
- **Combustion gas ka temperature zyada ho.**
- **Burning surface ke parallel gas ki velocity zyada ho** (isi ko erosive burning kehte hain, aage padhenge).
- **Rocket motor ki motion** - acceleration aur spin se grain par jo stress aata hai.

Composite propellant ka burn rate badhane ke 5 tareeke

Burn rate propellant ki composition ka function hota hai. Composite propellant me ise in tareeko se badha sakte hain:

1. Burning rate catalyst (modifier) daalo ya uski percentage badhao.

- Catalyst propellant ka sirf 0.1 se 3.0 percent hota hai. Ye bahut kam matra me daala jaane wala material hai jo burn rate ko badalta hai.

- Burn rate badhane ke liye ye mainly combustion zone me chemical reactions ko tez kar deta hai.

2. Oxidizer ke particles ka size chhota karo.

- Same mass ke oxidizer ke liye chhote particles ka total surface area bahut zyada hota hai.
- Isliye oxidizer aur fuel ka mixing aur reaction jaldi hota hai aur burn rate badh jaata hai.

3. Oxidizer ki percentage badhao.

- Ye is par depend karta hai ki fuel, oxidizer, binder aur formulation kya hai. Har case me same result nahi aata.

4. Binder ya plasticizer ki heat of combustion badhao.

- Binder ke 2 kaam hain: ingredients ko jodna, aur combustion me fuel dena.
- Plasticizer mainly mechanical aur processing properties theek karta hai, par kuch energy bhi de sakta hai.

5. Propellant me wires ya metal staples embed karo.

- Metal heat ko conduct karke aage ke propellant ko pehle se garam kar deta hai (preheat), jisse burning surface jaldi aage badhti hai.

> **Yaad rakho**
> Exam me composite ke 5 points ek line me: **catalyst, chhota oxidizer size, zyada oxidizer %, binder ki heat of combustion, embedded metal wire**. AP ka size kam ya percentage zyada karne par burn rate badhta hai (Lecture 3 ke graph ka point).

4. Grain design ki requirements

Grain design karte waqt kai cheezein ek saath satisfy karni padti hain. Slides me 5 requirements di hain:

1. Motor requirements (mission se aati hain)

- Ye mostly vehicle designers decide karte hain. Isme aate hain: total impulse, thrust-time curve aur uski tolerance, motor mass, storage aur operation ka ambient temperature limit, vehicle me available volume, motor ke center of gravity ki location, aur vehicle forces se aane wali accelerations (vibration, bending, aerodynamic loads).

2. Grain geometry

- Geometry motor requirements ke hisaab se choose hoti hai. Compact ho, available volume ka achha use kare.
- Burn surface vs time ka profile desired thrust-time curve se match kare.
- Erosive burning avoid ya predict ho sake. (Progressive burning wale kai motors thodi der ki erosive burning tolerate kar lete hain.)
- Bache hue unburned slivers aur burning ke dauran C.G. ka shift kam se kam ho.

3. Structural integrity

- Grain, liner aur insulator ko analyze karna padta hai taaki loading, acceleration ya thermal stress me grain fail na ho.
- Zyada stress ho to grain geometry badal ke stress kam kar sakte hain.

4. Internal cavity

- Perforations, slots, ports aur fins se bani cavity ka volume burning time ke saath badhta hai.
- Isliye in cavities ko resonance, vibration damping aur combustion stability ke liye check karna padta hai.

5. Processing

- Grain ki processing aur propellant fabrication repeatable, simple aur low cost honi chahiye, aur thermal stress sirf acceptable level tak aaye.

5. Ignition process

Ignition matlab ek signal (usually electric) milne se lekar motor ke apne aap chalne tak ki poori sequence. Isme ye hota hai:

- Signal milte hi igniter propellant jalta hai aur hot gas grain ki surface par jaati hai.
- Flame poori burning surface par fail jaati hai.
- Chamber gas se bharta hai aur pressure operating value tak pahunchta hai.
- Ye sab bina kisi gadbad ke hona chahiye. Gadbad matlab: overpressure, combustion oscillations, damaging shock waves, **hangfire** (delayed ignition), flame extinguishment aur chuffing.

Bade motors ko chhod ke baaki sab me ignition ek second ke hisse me complete ho jaana chahiye. Pressure bahut kam time me equilibrium value tak pahunch jaata hai.

Ignition ke 3 phases

Phase	Naam	Kya hota hai
I	Ignition time lag	Igniter ko signal milne se lekar tab tak, jab grain ka kuch hissa jal kar hot gas banane lage.
II	Flame-spreading interval	Grain ki pehli ignition se lekar poori burning area ke ignite hone tak ka time.
III	Chamber-filling interval	Chamber me gas bharne aur equilibrium chamber pressure aur mass flow tak pahunchne ka time.

Ignition tab successful maana jaata hai jab itni grain surface ignite ho chuki ho ki motor khud apna pressure operating chamber pressure tak badha le. Agar igniter kamzor hai to kuch surface thodi der jal kar bujh jaati hai. Slide ke hisaab se critical process burning surface ke upar gas-phase reaction me hota hai, jahan propellant vapors aur decomposition products, igniter gas ke products se interact karte hain.

Equilibrium pressure tak sahi pahunchna in 5 cheezon par depend karta hai

- Igniter ki characteristics: gas temperature, composition aur flow.
- Motor propellant ki composition aur grain surface ki ignitability.
- Igniter gas aur grain surface ke beech radiation aur convection se heat transfer.
- Grain par flame kitni tezi se spread hoti hai.
- Motor ke free volume ko hot gas se bharne ki dynamics.

Kisi particular motor grain ko ignite karne ke liye kitni aur kis type ki heat energy chahiye, ye igniter ke design parameters, khaas kar uske required heat output, ko seedha decide karta hai.

6. Propellant ignitability - 10 factors

Propellant kitni aasani se ignite hota hai (ignitability), ye in 10 cheezon se affect hota hai. Lecture 3 me jo explanation di hai wo saath me likhi hai:

#	Factor	Asar kaise hota hai
1	Propellant formulation	Kaun se ingredients hain, ye ignition behaviour decide karta hai.
2	Grain surface ka initial temperature	Temperature zyada = surface ko ignition point tak lane ke liye kam external energy chahiye, isliye ignite karna aasan.
3	Surrounding pressure	Pressure zyada = gas density zyada, gas-phase reactions tez, heat transfer behtar.
4	Heat transfer ke modes	Conduction, convection ya radiation, kaun sa mode kitna heat deta hai.
5	Grain surface roughness	Roughness se chhoti cavities aur zyada surface area bante hain jo heat trap karte hain, local hot spots banate hain aur convective heat transfer badhate hain.

#	Factor	Asar kaise hota hai
6	Propellant ki age	Purana hone par binder-oxidizer dewetting ya micro-cracking ho sakti hai jisse burning surface area bahut badal jaata hai.
7	Igniter gas ki composition aur hot solid particles	Igniter gas me hot solid particles (jaise zirconium ya boron oxides) hon to ye grain par takra kar bahut local aur strong conductive heating dete hain, jo sirf convective heating se zyada effective hai.
8	Igniter propellant aur uska initial temperature	Thanda igniter propellant dheere jalta hai, mass flow kam hota hai aur motor ignition late ho jaati hai.
9	Hot igniter gas ki velocity (grain surface ke relative)	Velocity zyada = convective heat transfer coefficient zyada, grain jaldi garam. Par bahut zyada velocity par local cooling (erosive cooling) ho sakti hai.
10	Cavity ka volume aur configuration	Cavity jitni badi ya jaisi hogi, utna hi gas ko bharne aur pressure banane me time lagega.

Slide 'Propellant ignitability curves' me heat flux vs ignition time ka graph hai: heat flux jitna zyada, ignition time utna kam.

7. Extinction ya thrust termination

Kabhi kabhi saara propellant jalne se pehle hi motor ko band karna padta hai. Ye cases slides me diye hain:

- Vehicle apni desired velocity tak pahunch gaya (ballistic missile apni fixed velocity par ya satellite apni orbit height par), ya precise total impulse cutoff chahiye.
- Safety ke liye, jab flight test vehicle test range ki safe boundary se bahar jaata dikhe.
- Multistage vehicle me stage separation ke time stages ki collision bachane ke liye (isme thrust reversal chahiye hota hai).
- Research aur development me, jab partially burned motor ko examine karna ho.

Ye kaam aise explosive devices se hota hai jo bahut predictable aur reproducible hote hain. Ye suddenly extra gas escape area khol dete hain (isse pressure girta hai) aur gas ko nozzle ki ulti direction me nikaal kar thrust ko neutralize kar dete hain.

Extinction ke common mechanism

- **Bahut tez depressurization:** nozzle throat area ko achanak bahut bada kar dena, ya extra gas escape ports jaldi khol dena.
- Sabse common technique: motor case ke forward end me exhaust ports achanak khol dena. Isse net thrust zero ho jaata hai ya ulta ho jaata hai. Aisa thrust reversal Minuteman aur Poseidon missiles ke upper stages me forward bulkhead ke ports se kiya jaata hai.

8. Inadvertent (anjaane me) ignition

Agar motor galti se ignite ho jaye to hot exhaust se jalna, aag, aas-paas ke motors ka ignite hona ho sakta hai. Agar motor bandha nahi hai to uska thrust use bahut tez ya erratic path par bhej dega, jisse bada damage hoga. Exhaust cloud toxic aur corrosive bhi hota hai. Galat ignition ke causes:

- Stray ya induced currents jo igniter ko activate kar dein.
- Electrostatic charging se sparks ya arc discharge.
- Aag lagne par motor ka bahar ka hissa itna garam ho jaye ki propellant temperature ignition point ke upar chala jaye.
- Impact: bullet penetration ya motor ko hard surface par gira dena.
- Lambe mechanical vibrations ki energy absorb hona jisse propellant overheat ho (jaise kharab sadak par transport).
- Nuclear explosions ki radiation.

9. Igniter hardware

Igniter propellant ka mass bahut kam hota hai (aksar motor propellant ke 1% se bhi kam) aur wo low chamber pressure par jalta hai, isliye uska Isp low hota hai. Isliye motor ke total impulse me igniter ka contribution negligible hota hai. Agar igniter forward end par laga ho to propellant surface par gas flow ignition me madad karta hai.

Igniter mounting options

Option	Kahan laga hota hai
Forward, Internal	Motor case ke andar, closed front dome par. Ye central cavity me nozzle ki taraf seedhi aag ki stream marta hai. Bade space boosters me sabse common setup.
Forward, External	Igniter ka main body front dome ke bahar laga hota hai, sirf uska nozzle ya port andar aata hai. Igniter hardware high-pressure chamber se alag rehta hai.
Aft, Internal	Rear me nozzle throat ke paas. Ye nozzle exit cone se ek support rod se tika hota hai. Motor ignite hone aur pressure banne ke baad poora igniter aur support assembly jaan boojh kar nozzle se bahar nikal diya jaata hai taaki exhaust path clear ho.
Aft, External	Rear dome ya nozzle housing ke bahar laga hota hai aur peeche se ignition gas ko cavity me upar ki taraf bhejta hai.

Igniter ke 2 basic aur sabse common types hain: **pyrotechnic igniter** aur **pyrogen igniter**.

(i) Pyrotechnic igniters

- Ye solid explosive ya propellant jaise energetic chemical formulations (aksar chhoti pellets, jinki burning surface badi aur burning time kam hota hai) use karte hain.
- Ye thode time ke liye bahut saari hot gas ya flame banate hain jisse main solid propellant ignite hota hai.
- Trade me inke naam hain: bag aur carbon igniters, powder can, plastic case, pellet basket, perforated tube, combustible case, jellyroll, string aur sheet igniters.

Type	Matlab
Bag igniter	Energetic material ek bag me bhara hota hai
Carbon igniter	Carbon-based container ya structure
Powder-can igniter	Material can jaisi enclosure me
Plastic-case igniter	Material plastic case me band
Pellet-basket igniter	Pellets ek basket jaisi structure me
Perforated-tube igniter	Chhed wali tube me material
Combustible-case igniter	Case khud jal jaata hai
Jellyroll igniter	Material rolled form me
String igniter	Material string ki shape me
Sheet igniter	Material sheet form me

Ignition sequence (pyrotechnic igniter)

Electrical signal -> Squib / primer -> Booster -> Main ignition charge -> Main propellant

- **Electrical signal:** initiator par electric signal lagta hai to wo ek chhota sensitive pyrotechnic material activate kar deta hai.
- **Squib / primer charge:** initiator me thoda sa sensitive powder pyrotechnic material hota hai (squib ya primer). Ye chhoti par intense heat aur hot gas banata hai.
- **Booster charge:** squib ki heat booster ko jalati hai. Booster me squib se zyada energetic material hota hai, isliye isse kaafi zyada heat aur gas milti hai.

- **Main ignition charge:** booster itni heat aur gas deta hai ki main ignition charge jal jaye, aur phir main propellant ignite hota hai.

Surface-bonded ya grain-mounted igniter: ye pyrotechnic igniter ka special form hai jo seedha solid propellant grain ki surface par laga hota hai.

(ii) Pyrogen igniters

- Pyrogen igniter ek chhota sa unit hota hai jisme rocket motor ke saare elements hote hain, aur ye bade rocket motor ko ignite karne ke liye use hota hai.
- Isme ek ya zyada nozzle orifices hote hain (sonic aur supersonic dono type) aur zyadatar conventional rocket motor grain formulation (kabhi kabhi main grain wali hi) use hoti hai.
- Pyrogen se grain ko heat mainly **convection** se milti hai, matlab hot gas grain ki inner surface ko touch karti hai. Pyrotechnic igniters me heat transfer high-energy **radiation** se hota hai. (Ye difference exam me puchh sakte hain.)
- Initiator aur booster charge ka design pyrotechnic igniters jaisa hi hota hai.
- Main charge ke reaction products motor grain ki inner surface par takrate hain aur motor ignite hota hai.

	Pyrotechnic igniter	Pyrogen igniter
Heat source	Solid explosive ya propellant pellets	Chhota rocket motor (apna grain aur nozzle)
Heat transfer	Mainly radiation	Mainly convection
Output	Thode time ke liye bahut hot gas ya flame	Nozzle se hot gas jet jo grain surface par takraata hai

Safe and Arm (S&A;) device

Motor galti se ignite na ho jaye, iske liye 2 tareeke hain: ek classical **safe and arm device**, aur dusra initiator ke andar hi safeguards design karna. S&A; device **SAFE position** me ignition do tareeko se rokta hai:

- **Electrical blocking:** initiator ka electrical connection toot jaata hai ya ground kar diya jaata hai.
- **Mechanical blocking:** ek barrier ya rotor aisi position me hota hai ki ignition path aligned nahi hota.

Jab device ko **ARM position** me ghumate hain to electric ignition circuit open ho jaata hai aur igniter ke booster aur main charge se flame propellant surface tak sahi se pahunch sakti hai.

Galat ignition ke energy sources: (1) static electricity, (2) electromagnetic radiation (jaise radar) se induced currents, (3) ground test equipment, communication apparatus ya vehicle ke nazdeeki electrical circuits se induced currents, (4) motor handling aur operations ki heat, vibration ya shock.

Electric initiators

Electric initiators ko squibs, glow plugs, primers ya headers bhi kehte hain. Ye ignition train ka sabse pehla element hote hain, aur agar sahi design ho to unintended ignition se bachane wala safeguard bhi ban sakte hain. Slides me 3 type diye hain:

Type	Kaise kaam karta hai
(a) Integral diaphragm type	Shock wave ke form me initial ignition energy diaphragm ke through acceptor charge ko activate karti hai. Diaphragm toot'ta nahi, integral rehta hai. Wahi principle metal case walls ya filament-wound cases ke metal inserts me bhi use hota hai, isliye combustion case me hole nahi karna padta aur wo sealed rehta hai.
(b) Header type (double bridgewire)	Simple glow plug jaisa. Initiator propellant charge me do high-resistance bridgewires dabe hote hain.
(c) Exploding bridgewire type	Chhota (0.02 se 0.10 mm) low-resistance bridgewire, usually platinum ya gold ka. High-voltage discharge lagane par ye explode ho jaata hai.

10. Mass flow relations

Burning surface se jo gas har second bani, wahi propellant mass generation rate hai:

$$\dot{m} = A_b \, r \, \rho_b$$

Jahan,

$\dot{m}$	=	Burning surface se banne wali gas ki mass flow rate, kg/s
A_b	=	Burning surface area, m^2
r	=	Burning rate, m/s
ρ_b	=	Solid propellant ki density, kg/m^3

Ye gas do jagah jaati hai: (1) chamber me jo gas store hoti hai uska change, aur (2) nozzle se bahar nikalne wali gas. Isliye:

$$A_b \, r \, \rho_b = \frac{d(\rho_1 V_1)}{dt} + \dot{m}_{nozzle}$$

- Pehla term (right side): chamber ke hot gas ke storage me change. Yaani chamber ka volume V1 badhta hai (grain jalne se cavity badi hoti hai) aur gas density ρ_1 badalti hai. ρ_1 = chamber hot-gas density.
- Aakhri term: nozzle se nikalne wala propellant flow rate.
- Start-up me chamber gas ke storage ka change important hota hai, par preliminary design me ise aksar ignore kar dete hain.
- Ab (burning area) constant rehta hai ya nahi, ye grain design par depend karta hai.
- Preliminary calculation me nozzle throat area At constant maan lete hain. Real hot firing me nozzle material ke oxidation, erosion aur abrasion se At badhta hai, par ye enlargement chhota hota hai (0.05 se 5%).
- Chamber gas volume V1 burn time ke saath measurable badhta hai.

Steady burning me chamber pressure ke liye sabse common relation:

$$p_1 = K \, \rho_b \, r \, C^* \qquad K \equiv \frac{A_b}{A_t}$$

Jahan,

p_1	=	Chamber pressure, Pa
K	=	Burning area aur throat area ka ratio (A_b / A_t)
C^*	=	Characteristic velocity, m/s
A_t	=	Nozzle throat area, m^2

Steady flow aur steady burning ke liye K (ya Ab) lagbhag constant rehna chahiye.

11. Burning rate aur pressure ka relation

Zyadatar propellants ke liye limited pressure range me burning rate ko chamber pressure ke function ke roop me likh sakte hain. Production-type propellants ke liye sabse common empirical equation (Saint Robert / Vieille law) ye hai:

$$r = a \, p_1^{\,n}$$

Jahan,

r	=	Burning rate, mm/s
a	=	Burning rate coefficient (propellant aur initial temperature par depend karta hai)
p_1	=	Chamber pressure
n	=	Burning rate exponent ya pressure exponent (dimensionless)

- Burn rate kisi propellant ke liye temperature aur pressure ki wide range me 3 ya 4 ke factor se badal sakta hai.
- Sabhi propellants ke liye burn rate lagbhag 0.05 se 75 mm/s (0.02 se 3 in./s) ke beech aata hai.
- Itna burn rate paane ke liye bahut chhote AP particles, burning rate catalysts, additives ya embedded metal wire ki zaroorat padti hai.
- AP particle size kam karne ya AP percentage badhane par burn rate badhta hai.

n ka matlab aur stability

- Zyadatar commercial propellants me **n = 0.2 se 0.6** hota hai.
- **n bada hone par** pressure me chhota sa change bhi burn rate (aur hot gas ki matra) me bada change laata hai.
- **n > 1.0** hone par chamber me jo bhi pressure disturbance ho wo amplify hota hai. Motor unstable ho jaata hai.
- **n 1.0 ke paas** pahunchne par burn rate aur pressure ek dusre ke bahut sensitive ho jaate hain aur chamber pressure kuch milliseconds me khatarnak tareeke se badh sakta hai.
- Isliye **stable operation ke liye n < 1** hona zaroori hai, aur jitna kam ho utna behtar (burn rate index for stable operation).

Plateau propellants: inme ek limited pressure range me burn rate lagbhag constant rehta hai. Ye isliye acche hain ki initial temperature ke change ka motor operation par asar kam ho jaata hai. Kuch double-base aur kuch composite propellants me ye plateau characteristic hoti hai.

Bade rocket motors ko ground-test firing se pehle kai ghante ek fixed temperature par **condition** kiya jaata hai taaki grain pura uniform temperature par ho, kyunki motor performance ko specified acceptable limits me rehna hota hai.

12. Ambient temperature ka burning rate par asar (T_b)

Propellant ka initial temperature Tb badalne par burn rate badalta hai. Slides ke numbers:

- Air-launched missile motors ke liye extremes usually **219 K (-65 F)** aur **344 K (160 F)**.
- Silo ya submarine launched motors ke liye **266 K (20 F)** se **300 K (80 F)**.
- Typical composite propellant motors me is range me chamber pressure me **20 se 35%** aur operating time me **20 se 30%** variation aa sakta hai.
- Graph (burning time vs chamber pressure) me +160 F, +70 F aur -65 F ke curves hain: garam grain me pressure zyada aur burn time kam, thanda grain me pressure kam aur burn time zyada.

Burn rate ki temperature sensitivity ko 2 temperature coefficients se express karte hain:

$$\sigma_p = \left(\frac{\partial \ln r}{\partial T_b}\right)_{p_1} = \frac{1}{r}\left(\frac{\partial r}{\partial T_b}\right)_{p_1}$$

$$\pi_K = \left(\frac{\partial \ln p_1}{\partial T_b}\right)_{K} = \frac{1}{p_1}\left(\frac{\partial p_1}{\partial T_b}\right)_{K}$$

Jahan,

σ_p	=	Temperature sensitivity of burning rate: fixed chamber pressure par propellant ke ambient temperature me per degree change se burn rate me change (1/K)
π_K	=	Temperature sensitivity of pressure: fixed K par propellant ke ambient temperature me per degree change se chamber pressure me change (1/K)
K	=	A_b / A_t

Agar n constant hai to $p_1 = (K \rho_b a C^*)^{1/(1-n)}$ se ye relation nikalta hai (slide me sirf ye bola hai ki π_K sirf σ_p aur n par depend karta hai):

$$\pi_K = \frac{\sigma_p}{1-n}$$

Matlab n jitna 1 ke paas jayega, π_K utna bada hoga, yaani temperature badalne par chamber pressure bahut jyada badlega.

Variable burning rate exponent n

Burn rate ke data se pata chalta hai ki n asal me constant nahi hota, balki p_1 ka function hota hai (aur isse indirectly Tb ka bhi). Isliye ambient temperature badalne par pressure excursions ko accurately predict karna shayad possible nahi hota. Constant-n case me π_K sirf propellant property (σ_p) aur n par depend karta hai.

13. Erosive burning

Erosive burning matlab burning surface ke upar combustion gases ke high-velocity flow ki wajah se propellant ke burning rate me **increase**.

- Ye solid rocket motors ke performance ko seriously affect kar sakta hai.
- Ye mainly port passages ya grain perforations me hota hai, jahan gas nozzle ki taraf behti hai.
- Ye tab zyada hota hai jab port passage ka cross-section area A, throat area At ke relative bada nahi hota. **Port-to-throat area ratio 4 ya usse kam** ho to erosive burning ka chance zyada hai.

14. Formula sheet (Unit 3)

Kya	Formula	Units
Burn time	$t_b = w / r$	w = web thickness, r = burn rate
Rocket equation	$\Delta V = V_e \ln(m_0/m_f)$	m/s
Mass generation	$\dot{m} = A_b r \rho_b$	kg/s
Chamber pressure	$p_1 = K \rho_b r C^*$, $K = A_b/A_t$	Pa
Burn rate law	$r = a p_1^n$	a ke units n par depend
Mass flow (thrust se)	$\dot{m} = F / (C_F C^*)$	kg/s
Mass flow (Isp se)	$\dot{m} = F / I_{sp}$	kg/s
Thrust	$F = C_F p_c A_t$	N
Characteristic velocity	$C^* = p_c A_t / \dot{m}$	m/s
Propellant mass	$m_p = \dot{m} t_b = \rho_p A_b L$	kg
Temp. sensitivity	$\sigma_p = (1/r)(dr/dT_b)$, $\pi_K = \sigma_p/(1-n)$	1/K

Solved Numericals - Unit 3

Slides me jitne bhi questions diye gaye the, un sab ko yahan collect karke solve kiya hai. Har question me order ye hai: **pehle formula, phir formula ke symbols ka matlab, phir given data, phir ek saath poori calculation** (numerator upar, denominator neeche).

Question 1

Ek propellant grain ki web thickness 40 mm hai aur wo 8 mm/s ke constant rate se jalta hai. Approximate burning time nikalo.

Formula: Burn time = web thickness / burning rate

$$t_b = \frac{w}{r}$$

Jahan,

t_b	=	Burning time, s
w	=	Web thickness, mm
r	=	Burning rate, mm/s

Ab, given data se hume pata hai ki

$$w = 40 \text{ mm} \qquad r = 8 \text{ mm/s}$$

Isliye,

$$t_b = \frac{40}{8}$$

$$t_b = 5 \text{ s}$$

$$\boxed{t_b \text{ (burning time)} = 5 \text{ s}}$$

Question 2

Ek single-stage rocket ko ΔV = 4500 m/s chahiye aur uska effective exhaust velocity V_e = 3000 m/s hai. Required initial-to-final mass ratio nikalo.

Formula: Rocket equation

$$\Delta V = V_e \ln\left(\frac{m_0}{m_f}\right) \qquad \Rightarrow \qquad \frac{m_0}{m_f} = e^{\Delta V/V_e}$$

Jahan,

ΔV	=	Rocket ko chahiye velocity change, m/s
V_e	=	Effective exhaust velocity, m/s
m_0	=	Initial mass (propellant ke saath), kg
m_f	=	Final mass (propellant jalne ke baad), kg

Given data se hume pata hai ki

$$\Delta V = 4500 \text{ m/s} \qquad V_e = 3000 \text{ m/s}$$

Isliye,

$$\frac{m_0}{m_f} = e^{4500/3000} = e^{1.5}$$

$$\frac{m_0}{m_f} = 4.48$$

$$\boxed{\frac{m_0}{m_f}\ (\text{required}) = 4.48}$$

Question 3

Ek end-burning rocket cylindrical double-base propellant grain use karta hai jiska diameter 200 mm hai. Ye 300 s tak 350 N thrust deta hai. Thrust coefficient 1.15 hai. Propellant ki density = 1500 kg/m^3, a = 4 mm/s, n = 0.5, C* = 1500 m/s. Nikalo: (i) grain ki length, (ii) nozzle ka throat diameter.

Given data

- Grain diameter, D = 200 mm = 0.2 m
- Thrust, F = 350 N
- Burning time, t_b = 300 s
- Thrust coefficient, C_F = 1.15
- Propellant density, ρ_p = 1500 kg/m^3
- Burning rate coefficient, a = 4 mm/s = 0.004 m/s
- Pressure exponent, n = 0.5
- Characteristic velocity, C* = 1500 m/s

Step 1: Propellant mass flow rate

Formula:

$$\dot{m} = \frac{F}{C_F C^*}$$

Jahan,

$\dot{m}$	=	Propellant mass flow rate, kg/s
F	=	Thrust, N
C_F	=	Thrust coefficient
C^*	=	Characteristic velocity, m/s

Isliye,

$$\dot{m} = \frac{350}{1.15 \times 1500} = \frac{350}{1725}$$

$$\boxed{\dot{m} = 0.2029\ \text{kg/s}}$$

Step 2: Total propellant mass

Formula:

$$m_p = \dot{m}\, t_b$$

Jahan,

m_p = Total propellant consumed, kg

t_b = Burning time, s

$$m_p = 0.2029 \times 300$$

$$\boxed{m_p = 60.87 \text{ kg}}$$

Step 3: Burning area (end-burning grain)

End-burning grain me burning area, circular end ka area hota hai.

$$A_b = \frac{\pi D^2}{4}$$

Jahan,

A_b = Burning area, m^2

D = Grain diameter, m

$$A_b = \frac{\pi (0.2)^2}{4}$$

$$\boxed{A_b = 0.031416 \text{ m}^2}$$

Step 4: Burning rate aur grain length - (i) ka answer

Formula: mass flow ki relation se burning rate, aur phir length = burning rate x burning time

$$\dot{m} = \rho_p A_b r_b \;\Rightarrow\; r_b = \frac{\dot{m}}{\rho_p A_b} \qquad\qquad L = r_b\, t_b$$

Jahan,

r_b = Burning rate, m/s

ρ_p = Propellant density, kg/m^3

L = Grain ki length, m

$$r_b = \frac{0.2029}{1500 \times 0.031416} = 0.004306 \text{ m/s} = 4.306 \text{ mm/s}$$

$$L = r_b\, t_b = 0.004306 \times 300$$

$$\boxed{L \approx 1.29 \text{ m}}$$

Check: L = m_p / (ρ_p A_b) = 60.87 / (1500 x 0.031416) = 1.292 m, same aaya.

Step 5: Chamber pressure

Formula: burning rate law

$$r_b = a\,p_c^{\,n} \;\Rightarrow\; p_c = \left(\frac{r_b}{a}\right)^{1/n}$$

Jahan,

p_c	=	Chamber pressure
a	=	Burning rate coefficient
n	=	Pressure exponent

Dhyan do: a = 4 mm/s tabhi lagta hai jab pressure MPa me liya jaye (slide me bhi yahi liya gaya hai). Isliye p_c MPa me aayega.

$$p_c = \left(\frac{4.306}{4}\right)^{1/0.5} = (1.0765)^2$$

$$\boxed{p_c = 1.159 \text{ MPa}}$$

Step 6: Throat diameter - (ii) ka answer

Formula: thrust equation se throat area, phir area se diameter

$$F = C_F\,p_c\,A_t \;\Rightarrow\; A_t = \frac{F}{C_F\,p_c} \qquad A_t = \frac{\pi d_t^{\,2}}{4} \;\Rightarrow\; d_t = \sqrt{\frac{4A_t}{\pi}}$$

Jahan,

A_t	=	Throat area, m^2
d_t	=	Throat diameter, m
C_F	=	Thrust coefficient

$$A_t = \frac{350}{1.15 \times 1.159 \times 10^6} = 2.626 \times 10^{-4} \text{ m}^2$$

$$d_t = \sqrt{\frac{4 \times 2.626 \times 10^{-4}}{\pi}} = 0.0183 \text{ m}$$

$$\boxed{d_t \approx 18.3 \text{ mm}}$$

Final answer

- (i) Grain ki length L = 1.29 m
- (ii) Throat diameter d_t = 18.3 mm

Question 4

Ek end-burning solid propellant rocket, jiska throat diameter 28 mm hai, missile ke sustainer ke roop me use hota hai. Use 5 kN ka constant thrust dena hai. Double-base propellant ka burn rate law r = 0.004 $P^{0.45}$ hai (P Pa me, r mm/s me). Specific impulse 1900 N.s/kg maan lo. Propellant ka C* = 1600 m/s aur density = 1540 kg/m^3 hai. Propellant grain ka diameter nikalo.

Given data

- Throat diameter, d_t = 28 mm = 0.028 m
- Thrust, F = 5 kN = 5000 N
- Burn rate law, r = 0.004 $P^{0.45}$ (P in Pa, r in mm/s)
- Specific impulse, I_{sp} = 1900 N.s/kg
- Characteristic velocity, C* = 1600 m/s
- Propellant density, ρ_p = 1540 kg/m^3

Step 1: Mass flow rate

Formula:

$$\dot{m} = \frac{F}{I_{sp}}$$

Jahan,

$\dot{m}$	=	Propellant mass flow rate, kg/s
F	=	Thrust, N
I_{sp}	=	Specific impulse, N.s/kg

$$\dot{m} = \frac{5000}{1900}$$

$$\boxed{\dot{m} = 2.632 \text{ kg/s}}$$

Step 2: Throat area aur chamber pressure

Formula: C* ki definition se

$$C^* = \frac{p_c A_t}{\dot{m}} \Rightarrow p_c = \frac{\dot{m} C^*}{A_t} \qquad A_t = \frac{\pi d_t^2}{4}$$

Jahan,

p_c	=	Chamber pressure, Pa
A_t	=	Throat area, m^2
d_t	=	Throat diameter, m

$$A_t = \frac{\pi (0.028)^2}{4} = 6.158 \times 10^{-4} \text{ m}^2$$

$$p_c = \frac{2.632 \times 1600}{6.158 \times 10^{-4}}$$

$$p_c = 6.839 \times 10^6 \text{ Pa}$$

Step 3: Burning rate

Formula: given burn rate law

$$r = 0.004\, P^{0.45}$$

Jahan,

r	=	Burning rate, mm/s
P	=	Chamber pressure, Pa

$$r = 0.004\,(6.839 \times 10^6)^{0.45} = 0.004 \times 1190.6$$

$$r = 4.762 \text{ mm/s} = 0.004762 \text{ m/s}$$

Step 4: Burning area aur grain diameter

Formula: mass flow relation se burning area, phir area se diameter

$$\dot{m} = \rho_p A_b r \;\Rightarrow\; A_b = \frac{\dot{m}}{\rho_p\, r} \qquad A_b = \frac{\pi D_p^2}{4} \;\Rightarrow\; D_p = \sqrt{\frac{4A_b}{\pi}}$$

Jahan,

A_b	=	Burning area, m^2
ρ_p	=	Propellant density, kg/m^3
r	=	Burning rate, m/s
D_p	=	Propellant grain diameter, m

$$A_b = \frac{2.632}{1540 \times 0.004762} = 0.3589 \text{ m}^2$$

$$D_p = \sqrt{\frac{4 \times 0.3589}{\pi}} = 0.676 \text{ m}$$

$$D_p \approx 676 \text{ mm}$$

Slide ke answer me difference

Lecture slide me is question ke aage ke steps me burning rate **r = 8.90 mm/s** likha hai aur final diameter **495 mm** aaya hai. Lekin 0.004 x (6.835 x 10^6)$^{0.45}$ calculator me daalne par r = **4.76 mm/s** aata hai, 8.90 nahi. Isi wajah se slide ka 495 mm galat lag raha hai; sahi value lagbhag **676 mm** hai. Exam me apna method aur calculator value hi likhna, aur sir se ek baar confirm kar lena.